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Matrix Theory

19 Determinants - II

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Dept. of AI, IIT Hyderabad
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Last class

- Geometric interpretation of Determinant (signed area/volume)

However

- Why is the formula for the determinant $(ad - bc)$, natural?

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- It is in the elimination process!

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- Jacobi (1840s) systematized the theory further and connected determinants to calculus

Determinant

Historical order: solve equations $\rightarrow$ notice a denominator $\rightarrow$ study that denominator $\rightarrow$ discover its properties $\rightarrow$ realize it measures volume

Teaching order: volume $\rightarrow$ axioms $\rightarrow$ formula $\rightarrow$ solve equations

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- let's define the determinant from the three axioms

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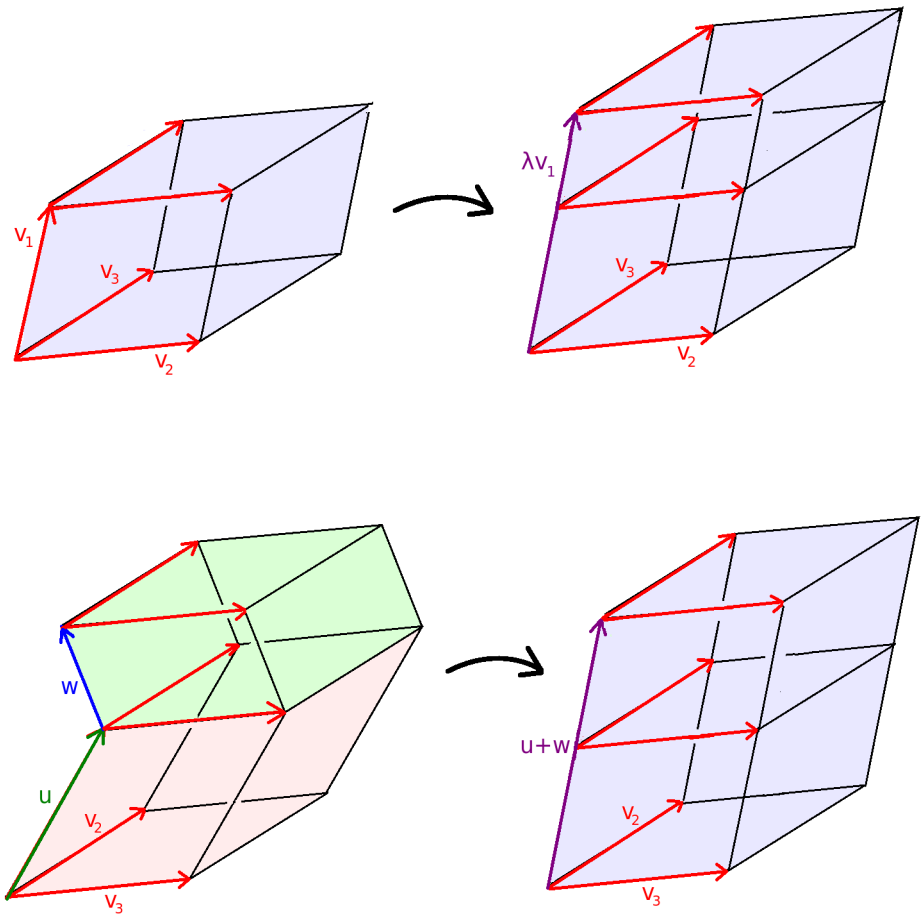
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  - A3 (Multilinear): $\det$ is linear in each column separately

Multilinearlity



volumes are equal by Cavalieri's principle
(imagine each stack as a stack of thin slices)

Credits: Nick Alger

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- Expand each column in the standard basis

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  $\det(A) = \sum_i \sum_j \sum_k a_{i1}a_{j2}a_{k3} \det(\begin{bmatrix} e_i & e_j & e_k \end{bmatrix})$

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- Only 6 survive (permutations of 1, 2, 3)
- $\det(e_i, e_j, e_k) = sign(\sigma)$ where σ is the permutation (i, j, k)

Determinant of a 3×3 matrix

$$\det(A) = a_{11}a_{22}a_{33} - a_{21}a_{12}a_{33} - a_{31}a_{22}a_{13} - a_{11}a_{32}a_{23} + a_{21}a_{32}a_{13} + a_{31}a_{12}a_{23}$$

Determinant of a $n \times n$ matrix

Leibniz formula :

$$\det(A) = \sum_{\sigma \in S_n} sign(\sigma) \cdot a_{\sigma(1),1} \cdot a_{\sigma(2),2} \cdots a_{\sigma(n),n}$$

Cofactor expansion (Laplace's Theorem)

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- Minor M_{ij}: Determinant of $(n-1) \times (n-1)$ submatrix removing row i and column j

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The value of $\det(A)$ is the same regardless of which row or column you expand along. Choosing a row or column with many zeros minimizes computation.

Determinant of Triangular matrices

det(upper or lower triangular) = product of diagonal entries
(can also be proved with row operations)

$\det A = \det A^T$

$$\det(A) = \sum_j a_{1j} \cdot (-1)^{(1+j)} \cdot \det(M_{1j}(A))$$

$$\begin{aligned}\det(A^T) &= \sum_j (A^T)_{j1} \cdot (-1)^{(j+1)} \cdot \det(M_{j1}(A^T)) \\ &= \sum_j a_{1j} \cdot (-1)^{(1+j)} \cdot \det(M_{j1}(A^T))\end{aligned}$$

Some more properties

- If two rows of A are equal, then $\det A = 0$ (sign change for row swapping)

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- Subtracting a multiple of one row from another row leaves the same determinant
- If A has a row of zeros, then $\det A = 0$

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- $\det A$ is the product of the pivots

Next

- Eigenvectors $Ax = \lambda x$